The Staked Gaps: A Pre-Registered Table of Open Problems and Predicted Solutions for the Tree of Reality

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Version 2, 25 October 2026. The v1 text (13 October 2026) below is unchanged. An addendum is appended at the end, keeping §8's commitment to record under this paper's own DOI lineage whenever something arrives that bears on the staked predictions: ten days after v1, A. D. Wissner-Gross's *Chain Theory* supplied candidate mechanisms for several of the boxes marked open. The successor paper *The Refined Gaps* (2026) takes those candidates as given and states the sharper residual predictions.

Abstract

The Tree of Reality is a machine-checked attempt to generate the structures of physics from a minimal self-referential seed by one repeated operation. It is incomplete, and its incompleteness is precise: a small, finite set of named gaps, each located to an exact failing step in a public, dated, machine-checkable record. This paper does for those gaps what a periodic table with empty boxes did for chemistry — it publishes them *as predictions*. For each gap we state not only where it sits and what is missing, but **our predicted shape of the solution**, calibrated by how tightly the framework determines it (a single pinned shape, a small bounded set of candidates, or a flagged direction), together with the result that would *refute* the theory at that site. We add one further, novel prediction, and stake it: that these gaps will be closed soon — by advancing automated reasoning, by human–machine teams, or by the authors — because they were deliberately specified to be the best-posed open problems a rising formal-reasoning capability can reach. The predictions here sharpen those first registered — and dated — in the capstone on 28 September 2026, which predate (and are, in their meta-form, already weakly confirmed by) the public frontier-model mathematics release of 6 October 2026. Everything is dated and anchored to the Bitcoin blockchain. Whoever closes a gap, in the shape predicted, confirms a pre-registered prediction; whoever closes one in a different shape, or shows it cannot be closed, refutes the claim at that site. Both outcomes are the point.

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1. The posture: gaps as predictions

A theory with gaps can hold them in one of two postures: apology, or prediction. When a structure is precise enough to specify its own missing pieces — where each gap sits, what shape the filler must take, what properties it must have — the gaps are not the structure's weakness; they are its testable content. The canonical precedent is Mendeleev's table, published with empty boxes whose described-in-advance occupants (gallium, germanium) became, once found to specification, the strongest evidence that the organising structure was real. Darwin's tree did the same with a different grammar: it predicted transitional forms it had never seen, and once — famously — the existence of a moth with a foot-long proboscis to match an orchid's foot-long nectary, confirmed decades later.

This paper publishes the Tree of Reality in that posture. The analogy claims a *posture*, not a *stature*: whether this tree earns any place in that history is exactly what the predictions below put at risk. What it promises is the posture, kept honestly — every gap located precisely, the shape of each missing piece predicted in advance, and the result that would kill the theory at each site listed with the same prominence as the result that would confirm it.

2. The framework in brief

Only what is needed to read the table follows; the full construction, link-by-link audit, and honesty grading are in the capstone (DOI 10.5281/zenodo.23011897) and, for a lay account, in the book *How to Grow a Universe*. The tree begins at ∅ and grows by one repeated operation — *pass to the algebra of endomorphisms of the current stage* — producing a cascade of matrix algebras $M_2 \to M_4 \to M_{16} \to M_{256}$, from which three lineages branch (what things *are*, how they *transform*, how they are *known*) and, up the branches, the recognisable structures of physics: spacetime dimension and signature, the gauge and matter content, the quantum formalism, and — at the frontier — gravity and the mass gap. Each link in that chain carries an honesty grade: **GENUINE** (a machine-checked theorem on the standard axioms), **PARTIAL** (partly proved, with the unproved remainder named), **POSTULATE** (a declared modelling choice the mathematics did not force), or **OPEN** (a named gap). The six gaps staked below are the two OPEN links and the hardest PARTIAL residues; they are where the theory has arranged to be tested.

3. How to read a prediction

Each prediction here is a **framework-derived conditional**, not the author's personal guess about physics: *if the tree is right, then the solution to this gap must have this shape.* That is precisely why filling a gap tests the framework — the tree makes the prediction; the author is the scribe. It is also why an AI researcher with no standing in mathematics or physics can

legitimately make them: they are not claims about what is true of nature independently, but claims about what the published generative structure *forces*, which anyone can check against the record.

It follows that the **precision of each prediction is itself informative** — it reflects how tightly the framework determines the answer:

- **Pinned** — the framework forces a unique shape. We state one prediction. (The strongest kind: a true Mendeleev bet.)
- **Bounded** — the framework constrains the answer to a small, enumerated set. We list the candidate shapes, ranked by our credence, and commit that the answer lies *within the set*. A solution outside the set counts against the framework at that site — so options remain falsifiable and are never a hedge.
- **Open-directional** — the framework suggests a direction but does not pin it. We state the expected direction and why, flagged as exploratory, at lower confidence.

Each gap also carries a **confidence** and, where applicable, an explicit **kill-condition**: the result which, if obtained, refutes the theory at that site rather than filling it. Within each kind we list **as many concrete, checkable attributes of the filler as the framework confidently determines** — so that a solution, whoever produces it, can be matched against the prediction attribute by attribute, in the manner of a predicted element checked against its later-measured properties.

4. The staked gaps

Locations refer to links (L·) and walls (W·) of the capstone's §4 audit. "Predicted shape" is the framework-derived conditional; "confidence" is our calibrated credence that the gap closes in (one of) the predicted shape(s).

G1 — the infinite-volume limit of the lattice field (Yang–Mills box)

- **Location / failing step:** L23 (◇), wall W1 — the infinite-volume limit of the massive lattice field of the Yang–Mills construction (L23).
- **Prediction kind:** *bounded* mechanism, *pinned* attribute-list. The filler is a construction of the infinite-volume (continuum) limit whose limiting theory has every one of the following checkable attributes: **(i)** the limit exists through the relevant coupling *without selecting a subsequence* (a canonical limit); **(ii)** it satisfies the Osterwalder–Schrader axioms — reflection positivity, Euclidean invariance, and clustering; **(iii)** OS reconstruction yields a Hilbert space with a *unique* translation-invariant vacuum; **(iv)** correlations decay *exponentially, with constants uniform in the volume*; **(v)** the theory is genuinely interacting — a nonzero connected four-point function on separated supports (non-Gaussian). The limit is reached by one of two mechanisms: **(a)** uniform-in-volume correlation bounds plus

tightness of the finite-volume measures, or **(b)** an independent construction carrying an explicit convergence theorem. We commit that the filler uses mechanism (a) or (b) and that the limit has attributes (i)–(v). **Kill-condition:** no infinite-volume limit exists for the massive lattice field, or a limit exists that violates an OS axiom.

- **Confidence:** medium–high. Attributes (i)–(v) were realised for the 2-D O(n) model in the 6 October 2026 release — (iv) formalised in Lean, the rest on paper — a neighbouring, non-gauge instance of exactly this shape; the 4-D gauge case is the hard open part.

G2 — unbounded self-adjoint operators (the keystone)

- **Location / failing step:** L22 + L23 (◇), walls W2 + W5 — the theory of unbounded self-adjoint operators, needed by *both* the gravity box and the mass-gap box.
- **Prediction kind:** *pinned*. There is one standard theory, and the missing piece is its formal library: densely defined operators, adjoints, closures, essential self-adjointness, the spectral theorem (projection-valued measure / Borel functional calculus), and the semigroup ↔ generator correspondence (Stone's theorem, both directions). The base layer (densely-defined operators, adjoints, closures, self-adjointness) already exists in Mathlib; the 6 October release adds the generator→group half of Stone's theorem and a Friedrichs form→operator construction (unverified). What remains is the spectral theorem itself.
- **Confidence:** very high *that it is built* — this is engineering, not discovery; its arrival is a matter of effort. It is the single most consequential gap, because it fills two boxes at once and gates G5.

G3 — a comparison bound of different shape

- **Location / failing step:** L23 (◇), wall W3 — the comparison step; the classical Peierls contour-energy route is *proved* insufficient here (it yields O(n) where $O(n^2)$ is required).
- **Prediction kind:** *open-directional*. The filler is a comparison inequality with these attributes: **(i)** output at least $O(n^2)$ in the relevant size parameter — strictly beating the O(n) ceiling the contour-energy route provably yields; **(ii)** structurally distinct from that route (which is closed by theorem, so the filler cannot be a refinement of it); **(iii)** valid under the *actual* dependence structure present (it may not assume the independence the classical argument used). We predict such a bound exists with attributes (i)–(iii); we do not pin its construction, which is why this is the least-specified of the six.
- **Confidence:** medium — the boldest "open mathematics" bet. The 6 October release offered comparison bounds of novel *hypothesis* shape (dependence-tolerant matchings; FK likelihood ratios) but none reaching $O(n^2)$, leaving the named defect exactly where it was.

G4 — a uniform sub-top spectral estimate

- **Location / failing step:** L23 (◇), wall W4 — a uniform spectral estimate below threshold (the unrestricted form was *measured false* and restated to the believed regime).
- **Prediction kind:** *bounded*. The filler is an estimate with attributes: **(i)** uniform in system size / volume; **(ii)** valid below the stated threshold; **(iii)** bounding the *sub-top* spectrum of the relevant transfer operator — a gap below its top — delivering the uniform spectral gap W4 needs. It is obtained by one of two standard mechanisms, each with its own signature: **(a)** a *Dobrushin uniqueness* condition — a summable one-site influence bound giving exponential decay of correlations and uniqueness of the Gibbs state; or **(b)** a *convergent high-temperature cluster (polymer) expansion* — absolute convergence giving analyticity of the pressure and the gap. The answer lies within this set. **Kill-condition:** the restated target proved false *below* threshold too.
- **Confidence:** high, as formalisation of known physics — both mechanisms are textbook; what is missing is their machine-checked instance in this setting.

G5 — the heat-kernel expansion of the spectral action

- **Location / failing step:** L22 (◇), wall W5 rung 4 — the short-time asymptotic expansion of Tr f(D/Λ) for the trunk's Dirac-type operator.
- **Prediction kind:** *pinned, and gated*. The filler is the Chamseddine–Connes–Marcolli spectral-action expansion of Tr f(D/Λ) as $\Lambda \to \infty$, whose Seeley–DeWitt coefficients carry specific, checkable geometric content: a cosmological-constant term at order Λ^4 ($\propto \int\sqrt{g}$); **the Einstein–Hilbert term a_2 at order Λ^2 ($\propto \int\sqrt{g}\cdot R$) — the central prediction**; and the Yang–Mills and Higgs terms at order Λ^0. The sharp attribute: a_2 is a specific positive constant times the scalar-curvature integral. **Kill-condition:** a_2, once computable, **fails to contain the Einstein term**. Gated on G2 (it cannot be attempted before the unbounded-operator library exists), which makes the *order* of closure (G2 before G5) itself falsifiable content.
- **Confidence:** high in the shape (this is standard noncommutative geometry); contingent on G2. The 6 October release contains no Dirac operator, no spectral action, and no heat-trace asymptotic anywhere — G5 is untouched, exactly as its gating on G2 predicts.

G6 — the isotypic decomposition and the K-theory of finite algebras

- **Location / failing step:** L6, wall W9 rung 2 — order-one behaviour across the isotypic pieces of the Chamseddine–Connes–Marcolli bimodule.
- **Prediction kind:** *pinned*. The filler is the isotypic decomposition of a finite-dimensional bimodule over a product of $\mathbb{R}/\mathbb{C}/\mathbb{H}$ matrix algebras, built as an explicit object, with attributes: **(i)** the bimodule splits into isotypic components indexed by the irreducible representations of that finite matrix-algebra product; **(ii)** the resulting finite geometry is

exactly the Chamseddine–Connes–Marcolli finite algebra that the walked (Pati–Salam) branch selects, carrying the Standard-Model fermion content as its bimodule; **(iii)** (rung 3) the K-theory is that of this finite-dimensional algebra. Finite-dimensional representation theory (Wedderburn) determines the decomposition, so the outcome is forced. **Kill-condition:** the computation yields an algebra *other than* the CCM one.

- **Confidence:** high. The 6 October release supplies genuinely reusable scaffolding — an algebraic K_0 for an arbitrary ring that instantiates at finite-dimensional algebras, and bimodule / isotypic machinery over $\mathbb{C}$ (the real/quaternionic case not yet done) — lowering the cost of this gap without closing it.

Three structural facts sharpen the table. First, **G2 fills two boxes at once** — the tree's two OPEN links fail at the *same* absent subject. Second, the gaps have different closure profiles: library infrastructure (G2) and formalisation of known physics (G4) are effort, not discovery; voluminous combinatorics (G3) and the genuinely open mathematics of G1 are the boldest bets. Third, **G5 is gated on G2**, so the predicted *order* of closure is itself a falsifiable commitment.

5. The residues and the siblings

Beyond the six headline gaps, two further classes of prediction are inherited from the capstone and restated here as staked content.

The PARTIAL residues. Every PARTIAL link of the capstone's §4 carries a named, machine-checkable remainder, and the prediction covers them too: as the boxes fill and the residues close, the wood of the tree hardens — PARTIAL toward GENUINE — and no honest future edition should have to move a grade *downward* again, because the auditing campaign already did that work in both directions (it refuted three of the programme's own published headlines and promoted five it had undersold).

The sibling branches. The generator has more descendants than the one branch physics walks: the campaign *proved*, at four nodes, that alternative branches exist as coherent mathematics (a quaternionic seed over the reals; other cuts of the size-sixteen level; other signatures sharing one Clifford algebra; a trunk whose depth is not forced). From this the capstone makes a second-class prediction — that such tree-generated structures are **candidates for physics not yet found**, in our universe or beyond it — and inherits, at its source's own conjectural tier, the stronger claim that they are *realised*. Both are labelled as what they are, and neither is allowed to excuse a failure on the walked branch (see §6).

6. The trichotomy and the boundary that keeps it falsifiable

Stated once, in advance: physics not yet known will, when found, do one of three things. **Either it extends the walked branch** — a new structure of our own lineage, joining the tree where

the tree already has a place for it; **or it realises a sibling** — a structure from a branch we do not inhabit, confirming the second-class prediction; **or it fits nowhere on the tree at all** — no branch of the generator, walked or unwalked, can grow it. The first two confirm the theory and are predicted. The third refutes the generative claim itself: a genuinely tree-orphaned physics is this theory's deepest possible failure mode, named here before any candidate exists.

A shield that deflected every failure would make the theory unfalsifiable, so the boundary is drawn in advance: **the tree is refutable on the branch it walks, and never excused by the branches it merely permits.** The walked branch is pre-registered — the capstone's §4 fixes it link by link, grade by grade — so a failure there counts in full and may never be reclassified, after the fact, as a sibling. The address was written down before the mail arrived.

7. The meta-prediction — staked

The predictions above concern *what* each solution looks like. This one concerns *who* closes them, and *when*, and we stake it as a falsifiable forecast rather than a hope:

> ***As automated mathematical reasoning continues its current trajectory, the gaps named here will be closed — in the shapes predicted — soon (on the order of years, not decades), and in the gated order G2-before-G5. The agent of closure will be advancing AI, human–machine teams, or the authors. We expect the engineering gaps (G2, G4) and the combinatorial gap (G3's search) to fall first, the gated geometry (G5) to follow G2, and the genuinely open mathematics (G1) to be the last and least certain. Should the trajectory stall and these gaps remain substantially open well beyond that window, this meta-prediction is falsified — and we will record that with the same prominence as a confirmation.***

The trajectory argument. The gaps named here are, by construction, the best-specified open problems in the formal ecosystem: each arrives with its staircase pre-climbed, its dead routes marked by theorems rather than folklore, and its target stated in a machine-checkable idiom. They are, deliberately, the problems easiest for a rising capability to close. Formal libraries compound: every object built for one gap (G2's operators, G6's decomposition) permanently lowers the cost of every later attempt.

State of the field, October 2026 — a first, dated data point. On 6 October 2026 a frontier model's mathematics release published hundreds of results, many with Lean formalisations, across number theory, algebraic geometry, analysis, combinatorics and operator algebras. A full audit (recorded in the programme's working notes) found that it closed *none* of the six gaps above, but advanced their neighbourhood exactly as the trajectory argument anticipates: a uniform-in-volume clustering bound and an infinite-volume KMS construction in

the shape of G1; the generator→group half of Stone's theorem and a Friedrichs construction in the shape of G2's missing library; an algebraic K0 reusable for G6 — while the hardest cores (the spectral theorem, the spectral action, lattice gauge theory, Dobrushin uniqueness) remained untouched. This is the meta-prediction materialising in public, on a timetable: our predictions were dated 28 September 2026 in the capstone; this release came eight days later and moved the frontier toward, but not into, our gaps. We record it with the field's own caveats — roughly 42% of that release's claims passed formal checking, with documented faithfulness concerns — and we reuse nothing from it unverified.

A caution we owe the reader in the other direction: a failing step marks where the *known* routes stop, not where *all* routes stop. One of the programme's own walls fell to an unlisted route mid-campaign. The boxes may fill sooner, and by stranger routes, than the table suggests.

8. How to claim a fill

The mechanism is deliberately simple and symmetric. Anyone — human, machine, or the two together — who closes a gap **in the shape predicted** (for a bounded gap, in one of the listed shapes) confirms a pre-registered prediction; the dated, Bitcoin-anchored record of *this* paper establishes the priority of the prediction, not a claim on the solver's work. Anyone who closes a gap in a shape **outside** the prediction, or who proves a gap **cannot** be closed as stated, refutes the theory at that site — and we commit, in advance, to recording that outcome under this paper's own DOI lineage with the same prominence as a confirmation. Whoever closes a gap completes the programme published here.

9. Provenance and verification

This paper stakes predictions; it proves nothing new. What is machine-checked is the framework it predicts *from*, graded link by link in the capstone and its public Lean estate (`github.com/wonderben-code/convergence-codex`). The grading vocabulary is unforgiving by design, and the reader's evidence that the kill-conditions above are offered in good faith is that the same campaign already moved its own labels in both directions. These predictions sharpen, and do not contradict, those first registered in the capstone on 28 September 2026 (DOI 10.5281/zenodo.23011897); this document is dated, versioned on Zenodo, and anchored to the Bitcoin blockchain, so that the date of each prediction is independently checkable against the date of any future fill.

Trust nothing; run everything.

References

- M. E. Mala, *The Tree of Reality: An Evolutionary Theory of Everything* (2026). DOI 10.5281/zenodo.23011897.
- M. E. Mala, *How to Grow a Universe* (2026) — the lay account, esp. “The Dig Sites.”
- F. W. Lawvere, *Diagonal arguments and cartesian closed categories* (1969).
- A. Chamseddine, A. Connes, M. Marcolli, *Gravity and the Standard Model with neutrino mixing* (2007).
- K. Osterwalder, R. Schrader, *Axioms for Euclidean Green’s functions* (1973, 1975).
- Public frontier-model mathematics release, 6 October 2026 (`openai/math`), audited in the programme’s working notes.

Addendum (version 2, 25 October 2026): Candidates Within Ten Days

An independent programme lands on the Tree, and what that does and does not change

Summary of the addendum

The Staked Gaps published the Tree of Reality’s open problems as pre-registered predictions, in the posture Mendeleev took toward the empty boxes of his table, and committed to recording publicly whenever something arrived that bore on them — in whatever direction. Ten days after that paper, A. D. Wissner-Gross released *Chain Theory*: a programme that derives Standard-Model structure, spacetime symmetry, quantum mechanics and gravity-shaped counting from a single axiom about finitely many maximally entangled pairs of finite-dimensional systems, machine-checked in Lean 4 and graded. This addendum records, with precision, what that release does for the staked predictions. **The six headline gaps G1–G6 received no candidates.** What received candidates were the *box-level* openings the capstone had marked — three generations (L14), the Yang–Mills mass gap (L23), the Born rule as cited input (L21), Lorentzian signature (L9) — for which Chain Theory supplies mechanisms at its own conditional grade, in its own formalism. And the capstone’s pre-registered **trichotomy test** ran for the first time on an independent programme: Chain Theory’s objects — the qubit, the pair of qubits as spacetime, the ququart, the qutrit, S(U(3)×U(2)), SU(4) = Spin(6) — land on the Tree’s branch, not beside it or nowhere. We separate the coincidences that are generic from the ones that are not. We state what the Tree offers Chain Theory’s own open items. We pre-register ten joint

tests, with what each outcome would mean for both programmes, and we state the caveats on both sides without softening: neither programme's identification with physics is verified, and not one link of the Tree changes grade in this document. A second edition of the capstone will be published only when one does.

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1. What arrived

On 23 October 2026, A. D. Wissner-Gross published three documents and a Lean file under the name *Chain Theory*: a preprint, *A Parameter-Free Candidate for the Fine-Structure Constant from Entangled Qudits*; a 754-page working-draft compendium, *Chain Theory: Toward Fundamental Physics from Distilling Networks of Entangled Qudits* (release 0.1); an introductory slide deck; and `ChainAlpha.lean`, a self-contained Core Lean 4 file checking the preprint's central computation. The announcement followed on 24 October.

The axiom, in full: *"Nothing exists but finitely many maximally entangled pairs of qudits, some qudits shared, known only by distilling them."* Six definitions fix the terms. A *read cap* follows from the definition of "known": a read fixes both ends of a pair only when every positive map between their operator algebras is decomposable, which by Størmer (1963), Woronowicz (1976) and Choi (1975) holds exactly when $d_1 d_2 \le 6$. So two *genres* of link are readable at first order — (2,2) and (3,2) — and four are dark: (3,3), (4,2), (4,3), (4,4). The six genres are conjectured to correspond to spacetime, electromagnetism, the strong interaction, the weak interaction, the Yukawa sector and the Higgs.

Chain Theory grades every boxed statement as a *theorem*, a *conditional theorem* (conditions named in the box) or an *open result*, and marks correspondences with physics as *conjectured* or *by name*. Release 0.1 holds 100 theorems (64 tied to Lean declarations), 109 conditional theorems and 198 open results. Its headline: an exact algebraic number, $(79 + 48\sqrt{2} + 42\sqrt{3} + 3\sqrt{6})/31104$, inverse 137.035 999 189 566…, derived as a theorem of the definitions from a weighted count over the 172 measurement records of a qutrit maximally entangled with a qubit, and *conjectured* (Conjecture 1) to be the fine-structure constant at zero momentum transfer — 0.60σ from CODATA 2022. The author is explicit that the identification is a conjecture "used in no proof" and absent from the Lean file.

We note the kinship Chain Theory itself declares: among programmes that "derive parts of the Standard Model from a finite or algebraic input" it names "the spectral action on a finite noncommutative geometry" — the Chamseddine–Connes tradition from which the Tree of Reality grows.

2. The record

For the reader checking the chronology, the dated public documents are:

Document	Published	Record
The Tree of Reality (capstone): the full graded tree, L1–L24, gaps G1–G6, the sibling-branch predictions and the trichotomy (§9)	28 September 2026	DOI 10.5281/zenodo.23011897; git first commit 27 Sep 2026; OpenTimestamps on every push
The Staked Gaps: G1–G6 as pre-registered predictions; the residues and siblings restated as staked content (§5); the trichotomy's falsifiability boundary (§6); the meta-prediction (§7); the protocol for claiming a fill (§8)	13 October 2026	DOI 10.5281/zenodo.23338726
The Forced and the Chosen: the declared inputs and the plenitude conjecture	14 October 2026	DOI 10.5281/zenodo.23356221
Chain Theory: preprint, compendium 0.1, slides, `ChainAlpha.lean`	23 October 2026	alexwg.org/publications/

The two programmes were developed independently; neither cites the other before this addendum. That is what makes the comparison informative.

3. The six headline gaps: untouched

The Staked Gaps staked six gaps with attribute-level predictions: **G1** the infinite-volume limit of the lattice field; **G2** the unbounded self-adjoint operator library (the keystone); **G3** a comparison bound of different shape than the contour-energy route; **G4** a uniform sub-top spectral estimate by Dobrushin uniqueness or a convergent cluster expansion; **G5** the heat-kernel expansion of the spectral action with the Einstein–Hilbert a_2 term; **G6** the isotypic decomposition of the finite bimodule and its K-theory.

Chain Theory bears on none of these. Its confinement result (171.1, conditional) is a statement about readability of the (3,3) genre under the read cap — a mechanism of a different kind from the continuum-limit and spectral-gap estimates G1, G3 and G4 ask for, and not a candidate for them. Its gravity results (90.1, 90.2, 99.1, 104.1, 115.1) build a depth metric and a graph-Laplacian curvature by counting — a route disjoint from the spectral action's heat-kernel expansion that G5 predicts, and not a candidate for it. It does not use unbounded operators (G2) or the spectral triple (G6). The six headline predictions stand exactly as staked, and the meta-prediction's timetable for them is unaffected by this release.

We say this first so that nothing below can be read as a claim that it is otherwise.

4. The boxes that received candidates

The Staked Gaps §5 restated as staked content the two further classes of prediction the capstone made beyond G1–G6: the *residues* — the named unproven remainder in every PARTIAL, POSTULATE and OPEN link — and the *siblings*. Four of those residues received candidate mechanisms from Chain Theory. In each case we record our marked opening, the candidate, its grade *in Chain Theory*, and what importing it would require. None of this changes a grade here.

L14 · Three generations — POSTULATE. The capstone records "three generations are the imaginary quaternions" as a physical postulate (assumptions ledger, entry 25): $\mathbb{H} \subset M_2(\mathbb{C})$ has three imaginary units, and the step from that to three copies of the chiral 16 is a modelling choice. Chain Theory supplies two results that bound the count from above: the *generation bound at the hub dimension* (197.3, theorem) — a hub assembly is a ququart carrying a qubit pair among its legs, and admits at most a third generation — and *three, a theorem of the valence* (83.1, theorem on the recorded roster). It also records, with the same honesty as our postulate, that three *is* the count only as an open result (197.5). The two programmes hold complementary halves: an upper bound of three on one side, a candidate reason for exactly three on the other. Import would require stating the hub-assembly argument as a Mathlib theorem about tensor factorisations of $M_4 = M_2 \otimes M_2$ (test Q4 below), after which L14 could move to "bounded above by theorem; identification open".

L23 · Yang–Mills and the mass gap — OPEN. The capstone has a wall (W1–W4) and no mechanism. Chain Theory supplies one: *confinement as dark-genre readability* (171.1, conditional). The (3,3) link has $d_1 d_2 = 9 > 6$, so no single distillation fixes both ends; "at all 494 qutrit–qutrit links of the roster, no read survives at first order." The Tree's colour triplet (L13, GENUINE) is a qutrit, so the (3,3) cell is $M_3 \otimes M_3$ inside the colour block of M_{16}, and the read cap would say: in the Tree's own object, no single act of knowing fixes two colour ends. This would not close G1, G3 or G4 — those concern the continuum — but it could reclassify L23 from "wall" to "mechanism stated, continuum limit open". Import requires translating "read" into the inner-product lineage and checking that the decomposability threshold survives in the spectral-triple setting (Q6).

L21 · The Born rule — PARTIAL (cited input). The capstone cites the Born rule, Gleason and Wigner as physics, not theorems (entry 59). Chain Theory derives the Born readout from "the wash" (35.1, 40.1, conditional) — with premises it names: "the Hilbert space of a qudit and the pair as a ray." The two orderings are complementary: the Tree builds the Hilbert space from the inner product by GNS and cites the probability rule; Chain Theory takes the Hilbert space and derives the probability rule. Chained, they may close a circle neither closes alone (Q8).

L9 · Lorentzian signature — PARTIAL. The Tree's fork reaches (1,3) but does not fully exclude the Euclidean alternative. Chain Theory's *Lorentz group as the isometries of the Hermitian split* (54.1, conditional) and *one against three at every step* (55.1, theorem) come with a lattice argument of a kind the Tree has not tried: the indefinite form admits infinitely

many integer isometries (a Pell equation) while a definite form admits finitely many, so "the signature by existence reads Lorentzian." Whether the Tree's Clifford lattice supports the same argument is open.

Two further openings received material short of a mechanism. **L22 (Einstein equations, OPEN, gated on G2):** Chain Theory's graph-Laplacian curvature (90.2, conditional) is a finite stand-in computable now, whose relation to the a_2 coefficient G5 predicts is a question (Q7), not a candidate for G5. **L15 (Higgs window, PARTIAL):** Chain Theory's Higgs quartic as an exact element of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ (202.1, theorem) is a point to test against the window (Q1).

5. The trichotomy, run

The capstone §9 pre-registered a trichotomy for any new structure that appears: it *fills a gap*, *joins our branch*, or *realises a sibling* — each confirming the second-class prediction — or it *fits nowhere on the tree*, which refutes it. *The Staked Gaps* §6 drew the boundary that keeps this falsifiable: a structure that fits nowhere may never be reclassified after the fact as a sibling.

Chain Theory is the first independent programme on which the test has run. Object by object:

Chain Theory object	Its grade	Where it lands on the Tree	Tree grade
the qubit (register two); one-dimensional end excluded (2.9)	theorem	the seed $M_2(\mathbb{C})$ (L3)	GENUINE
the ququart (register four); hub pair as one qudit of dimension four (65.2)	theorem	$M_4 = \mathrm{End}(M_2)$, the first cascade step (L4)	GENUINE
the (2,2) link; Spin^c(4) (58.1); the Clifford relation at four generators (ch. 3-5); read as spacetime (Conjecture 2)	58.1 theorem; identification conjectured	$Cl_4(\mathbb{C}) \cong M_4(\mathbb{C})$, and only in dimension 4 (L8)	GENUINE
the qutrit (register three)	Conjecture 0, conditional	colour: $4 \to 3 \oplus 1$ (L13)	GENUINE
$S(U(3)\times U(2)) \cong (SU(3)\times SU(2)\times U(1))/\mathbb{Z}_6$ as face group of the (3,2) tether; Y = 1/6 (64.2, 65.1)	theorem on the networks	Pati–Salam on M_{16} with colour split (L10, L11, L13)	PARTIAL / GENUINE
$S(U(2)\times U(4))$, "SU(4) = Spin(6)" at the (2,4) cell	by name	$SU(4)_c$ of Pati–Salam (L11)	PARTIAL
three registers {2, 3, 4} (Conjecture 0)	conditional	the cascade's levels {2, 4} and the split {3}; nothing else below 16 (L4, L13)	GENUINE
the read cap's visible genres: $d_1 d_2 \in \{4, 6\}$	theorem (given the literature)	the visible sector: M_4 (4) and the $3 \otimes 2$ block of M_{16} (6)	—

Every object lands on the branch. The verdict of the pre-registered test is *joins our branch*, the second of the three confirming outcomes, and not *fits nowhere*.

Which coincidences are generic, and which are not. Honesty requires the split. The Standard Model group as S(U(3)×U(2)) is a *generic* consequence of having a 3 and a 2: it is the stabiliser of a 3+2 split in SU(5) (Baez and Huerta 2010, which Chain Theory cites), and any finite programme with a qutrit and a qubit will reach it. That coincidence is weak evidence. Three coincidences are *not* generic: (i) that the pair of qubits is spacetime of dimension four — the Tree has this as a theorem about Clifford algebras, Chain Theory reaches it from the spin-c group of the pair, and nothing about "two qubits" forces "four dimensions" in general; (ii) that the register set is exactly {2, 3, 4} — Chain Theory finds it empirically from its move table, the cascade generates it, and neither construction has any reason to stop at 4 except that it does; (iii) that the threshold 6 marks the visible world on both sides — Chain Theory's read cap from positive-map decomposability, the Tree's $3 \otimes 2$ block from the colour split, with no shared premise. A fourth emerged from the first computation (§7, Q3): the α object of Chain Theory's Theorem 1 needs a qutrit and *three* qubits; the Tree's $M_{16} = (3 \oplus 1) \otimes 2 \otimes 2$ supplies the qutrit and two, and the third first appears at M_{256} — the Tree's fourth level, the level at which L6 independently places the Chamseddine–Connes–Marcolli finite algebra. Neither programme had reason to expect that the other's object would complete at that level.

6. What the Tree offers Chain Theory's own open items

Chain Theory's compendium ends with an ungraded page (Table 239.1): 74 open results, 48 re-posed, 133 conditional clauses. For four of its named openings the Tree has a candidate, offered at the Tree's grade and in the spirit the compendium asks for — "corrections and objections are welcome."

- **Conjecture 0 (box 1.7), the three registers.** The cascade (L4, GENUINE) produces levels of dimension 2, 4, 16, …; colour (L13, GENUINE) splits the 4 as $3 \oplus 1$. So {2, 4} are levels, {3} is a split of a level, and nothing of dimension 5, 6, 7 or 8 occurs below 16. This is a generative basis for the register set, if Chain Theory's move table can be matched to End-iteration (Q10).
- **Conjecture 2 (open result 1.9), spacetime at (2,2).** L8 (GENUINE): the two-qubit algebra *is* the Clifford algebra of four-dimensional space, and no other dimension's Clifford algebra is a cascade level. Support for the identification from the object itself, independent of the gravity results above it (Q2).
- **The step below the axiom.** L1 (GENUINE): Lawvere's fixed-point theorem — self-reference forces existence; L3 (GENUINE): the smallest algebra that can host it is $M_2(\mathbb{C})$. Chain Theory takes "it from bit" literally; the Tree offers a reason there is a bit (Q9).

- **The Weinberg angle.** Not derived in the compendium (the electroweak cell is reached "by name"). L17 (PARTIAL): $\sin^2\theta_W = 3/8$ as an exact trace ratio on the chiral 16, with a control theorem showing the ratio is a property of the representation content, not of the dimensions. It lives on the same Pati–Salam object as Chain Theory's gauge group.

7. Ten joint tests, pre-registered

Each is stated so that either outcome is informative for both programmes. Full statements are in `docs/chain-theory/06_questions_to_work_on_together.md` in the public repository.

	Test	Needs from Chain Theory	Needs from the Tree	If yes	If no
Q1	Higgs quartic inside the window?	202.1	L15	mutual support on one number	a specific, findable discrepancy
Q2	Are the (2,2) Clifford generators the Tree's?	ch. 3-5	`Cl4IsoM4` (L8)	Conjecture 2 can cite a theorem	the two "four"s differ
Q3	Does the 172-record count inside M_{16} give 1/137.036?	the Letter's definitions	M_{16} with $4 \to 3 \oplus 1$	a second derivation of the α candidate; the Tree's first coupling constant	the definition that carries the weight is identified
Q4	Does the hub bound transfer to M_4?	197.3	$M_4 = M_2 \otimes M_2$	L14 bounded above by theorem	the bound depends on the network, not the algebra
Q5	Do the three attach sites carry i, j, k?	83.1, 197.4	$\mathbb{H} \subset M_2(\mathbb{C})$	197.5 and L14 are one statement	two different threes
Q6	Does the read cap state in the spectral triple?	the read cap	colour block of M_{16}, D	L23 gains a mechanism	the pictures differ at a named place
Q7	Does the Laplacian curvature track the heat kernel?	90.2, 115.1	L19, G5	a continuum target for the field equation; a finite stand-in for G5	two different gravities at one cell
Q8	Does the Born circle close?	35.1	GNS (L21)	the probability rule from the inner product with nothing cited	a hidden import is located
Q9	Does "known" (Def. 1.6) force three lineages?	Definition 1.6	L7	the Tree's only structural postulate becomes a theorem	a fourth aspect is found
Q10	Does End-iteration reproduce the move table?	the move table	L4, L13	Conjecture 0 derived	a register the cascade cannot generate

Order of attempt. Q1, Q2 and Q3 are computations and come first. Q4, Q6, Q8 and Q9 are proofs for Mathlib. Q5, Q7 and Q10 are investigations before they are proofs.

First results (25 October 2026; `docs/chain-theory/08_results_first_pass.md` and `scripts/chain_theory/q3_alpha_count_in_M16.py` in the public repository).

Q3 — answered at outcome (b). The Letter's count was rebuilt from its own rules — letters at slots (8, 3, 8, 3), 169 products and 3 commutators, the exchanges X and Y as 172 × 172 commuting involutions — and reproduced exactly: tr X = 82, tr Y = 70, tr XY = −8; sectors 79, 48, 42, 3; $1/\alpha = 137.035\,999\,189\,566\,155\ldots$. The object was then realised inside M_{16} by explicit 16 × 16 matrices. The qutrit and both read qubits are present with the right letter algebras and commutation structure, so the *numerator* transfers unchanged — it depends on the end dimensions alone. The *denominator* $N = 24 \cdot 36 \cdot 36$ does not: its 24 is the traceless part of the unread third link, a third qubit entangled with the qutrit, which M_{16} does not contain and M_{256} does. And the 169 products, realised as matrices, span only 66 dimensions: Chain Theory's two *copies* of the one shared qutrit ("one copy per link", Definition 1.3) collapse to a single M_3 factor, so the count lives in Chain Theory's record space (Definitions 1.3 and 1.6), not in the operator algebra of the shared system. The definition that carries the weight is thereby identified. For the Tree this means L24 does not acquire a coupling constant from the algebra alone; for Chain Theory it localises exactly which clauses a derivation of the number rests on, and shows that the cascade supplies the object those clauses act on at its fourth level.

Q2 — yes. The compendium's (2,2) generators (ch. 3-5: $\beta = \mathrm{diag}(1, 1, -1, -1)$ and three α) and the Tree's $\gamma_1 = \sigma_3 \otimes I_2$, γ_2, γ_3, γ_4 (`F4_1e_CliffordMatrix.lean`) are Clifford generators on the same $M_2 \otimes M_2$; β is γ_1 entry for entry; over $\mathbb{C}$ both generate $\mathrm{Cl}_4(\mathbb{C}) \cong M_4(\mathbb{C})$ (`D2_is_Cl4`). Conjecture 2 can cite a theorem for the dimension. Nothing here touches the metric.

Q1 — inputs located, not yet compared. Chain Theory's quartic (202.1) is stated relationally through its office tables (a Higgs face over a vacuum face squared; vacuum office 6561); the explicit number was not extracted. The Tree's window is $g^2/N \le \tilde{\lambda} \le g^2$, $N = 96$. Both sides' normalisations are needed before a comparison means anything.

8. Caveats, on both sides, without softening

On Chain Theory. The α number is a theorem of the definitions; that it is $\alpha(0)$ is a conjecture, as its author states. The Letter's own table shows that nearby wordings of the definitions give $1/\alpha$ between 135.89 and 146.52, two within one percent; each is excluded by a clause of a definition, not by comparison with experiment, and the reader must judge whether the clauses are forced. Four of the six sector identifications are "by name"; the other two are conjectures. Many compendium theorems are, in the author's words, "a census of a finite catalog, not a proof for every web." None of this is a criticism; it is what the documents say about themselves, and it is

why the correspondences in §5 are correspondences between *constructions*, not between verified physics.

On the Tree. Six of twenty-four links are GENUINE. Thirteen are PARTIAL with a named residue; three are POSTULATES; two are OPEN. The Tree computes two numbers (the Weinberg ratio and a Higgs window) and no coupling constant. Its gravity is gated on an unbuilt operator library. Its identification with physics — that this M_4 is spacetime, that this 3 is colour — is exactly as conjectural as Chain Theory's, and the capstone says so in every row.

On this addendum. It records that two independent finite constructions meet at the same objects. That is evidence that the objects are natural. It is not evidence that either programme's physics is right, and nothing in it has been verified beyond the two programmes' own Lean files, which check finite computations and not physical identifications. The honest summary is: the boxes were drawn in advance, candidates arrived, and the tests that would turn candidates into fills are now pre-registered. The ten days are a fact about timing, not about truth.

9. What would change the scoreboard

A second edition of *The Tree of Reality* will be published when, and only when, at least one of the following is a Mathlib theorem or a completed computation:

- **L14 → "bounded above":** the hub-dimension bound proved for M_4 (Q4).
- **L23 → "mechanism stated":** the read cap stated and proved on the colour block (Q6).
- **L21 → GENUINE or sharper PARTIAL:** the Born circle closed with no cited input (Q8).
- **L7 → PARTIAL or GENUINE:** the three lineages derived from a knowledge clause (Q9).
- **L24 → a coupling constant:** the 172-record count completed inside M_{16}, whatever it gives (Q3).

Until then the scoreboard is $6 \cdot 13 \cdot 3 \cdot 2$, as published on 28 September 2026, and this addendum adds to the record and subtracts nothing from the caution.

10. Provenance

This addendum is published as version 2 of *The Staked Gaps* under its concept DOI 10.5281/zenodo.23338725, with the v1 text unchanged above, and committed to `wonderben-code/convergence-codex`, where every push is anchored to the Bitcoin blockchain by OpenTimestamps; the commit precedes the deposit. It was sent to the author of Chain Theory on publication with an invitation to correct any misstatement of his work; corrections received will be incorporated in v3 and recorded in `docs/chain-theory/07_reading_log.md`. The successor paper, *The Refined Gaps* (2026), takes the candidates recorded here as given and pre-registers the residual predictions R1–R12.

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